

Analysis

Human papilloma virus infection drives unique metabolic and immune profiles in head and neck and cervical cancers: implications for targeted therapies and prognostic markers

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Abstract

Human papillomavirus (HPV) is a key driver of head and neck squamous cell carcinoma (HNSCC) and cervical squamous cell carcinoma (CESC). Yet, these cancers exhibit distinct molecular and clinical features influenced by HPV status. This study utilizes RNA sequencing data from The Cancer Genome Atlas (TCGA). It employs bioinformatics tools, including DESeq2 for differential gene expression, CIBERSORT for immune profiling, and Kaplan–Meier survival analysis to investigate these differences. Differential expression analysis revealed distinct molecular signatures, with HPV-positive tumors enriched in immune-related pathways such as cytokine–cytokine receptor interactions. In contrast, HPV-negative tumors exhibited upregulation of metabolic pathways, including PPAR signaling. Metaflux analysis further demonstrated contrasting metabolic profiles: HPV-positive tumors showed increased glycolysis and oxidative stress regulation, whereas HPV-negative tumors were characterized by elevated amino acid and nucleotide metabolism. Immune profiling highlighted more significant CD8+ T-cell infiltration in HPV-positive tumors, while HPV-negative tumors were predominantly associated with macrophages, suggesting differing tumor immune environments. Survival analysis identified CXCL11 and STAT1 as potential prognostic biomarkers, with lower expression correlating with poorer survival in both cancers. These findings provide an integrated perspective on the molecular, metabolic, and immune differences associated with HPV status, offering insights into potential therapeutic strategies.

Keywords Human papillomavirus (HPV) · Head and neck squamous cell carcinoma (HNSCC) · Cervical squamous cell carcinoma (CESC) · Gene expression · Metabolic activity · Biomarkers

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1 Introduction

Cancer remains one of the most significant global health challenges, with an estimated 19.3 million new cases and 9.6 million deaths worldwide in 2022 [1]. This reflects the widespread impact of various types of cancer, including head and neck squamous cell carcinoma (HNSCC) and cervical squamous cell carcinoma (CESC). The HNSCC affects areas such as the mouth, throat, and larynx and is particularly common in regions where tobacco and alcohol use are prevalent [2]. It accounted for over 900,000 new cases globally, making it one of the most frequent cancers worldwide in 2022 [3]. The prognosis for HNSCC patients is challenging, as many cases are diagnosed at later stages when treatment options are more limited [4]. Additionally, risk factors such as human papillomavirus (HPV) infection and exposure to carcinogens like tobacco smoke and alcohol play a significant role in the development of these cancers [5]. On the other hand, CESC, which arises from the cervix, remains a major health issue for women, especially in low- and middle-income countries where access to screening and vaccination programs is limited [6]. In 2022, CESC led to nearly 600,000 new cases and around 340,000 deaths globally [7]. Despite these challenges, there have been significant strides in cancer prevention, particularly for CESC, with the widespread use of the HPV vaccine showing promise in reducing incidence rates.

HPV is a key player in developing both types of cancers [8]. It is a sexually transmitted virus that infects epithelial cells in the skin and mucous membranes [9]. There are more than 200 types of HPV, but the high-risk strains, such as HPV-16 and HPV-18, are most commonly associated with cancer development [10]. HPV infection leads to the integration of viral DNA into the host genome, producing viral proteins like E6 and E7 [11]. These proteins disrupt normal cell cycle regulation by inactivating tumor suppressor proteins such as p53 and retinoblastoma (Rb) [12]. This inactivation allows infected cells to bypass the usual growth control mechanisms, leading to the accumulation of genetic mutations and the eventual development of cancer [13]. In the case of CESC, persistent HPV infection is the primary cause of cervical cancer, with HPV being responsible for approximately 70% of all cervical cancer cases globally [14]. In HNSCC, HPV infection, particularly with HPV-16, has been linked to oropharyngeal cancers, a subset of head and neck cancers [15]. While these cancers were once primarily associated with tobacco and alcohol use, the increasing prevalence of HPV-associated HNSCC shows the growing importance of the virus in cancer epidemiology [16].

HPV-positive and HPV-negative cancers differ not only in their etiology but also in their biological characteristics, prognosis, and response to treatment [17]. These cancers tend to have a better prognosis compared to their HPV-counterparts [18]. Studies have shown that patients with HPV-positive CESC generally have higher survival rates, mainly due to the distinct nature of HPV-induced carcinogenesis [19]. HPV-positive cancers tend to have a more immune response, which may contribute to the ability of body to control tumor growth [20]. Additionally, HPV-positive tumors are often less aggressive, with slower progression and a better response to treatments such as radiation therapy [21]. In contrast, HPV-negative cancers often have a poorer prognosis, as they tend to be more aggressive, with a higher likelihood of metastasis and resistance to treatment [22]. The 5-year survival rate for patients with HPV-positive oropharyngeal cancer can exceed 80%, whereas for HPV-negative oropharyngeal cancer, the survival rate drops to around 50% [23]. The immune landscape in HPV-negative cancers is often more suppressive, with higher levels of macrophage infiltration and immune evasion mechanisms [24]. This difference in immune composition is one reason that HPV-negative cancers are harder to treat and generally have lower survival rates [25]. Moreover, HPV-negative HNSCC is frequently associated with a history of tobacco and alcohol use, both of which are established carcinogens that exacerbate cancer progression [26].

Moreover, distinguishing between HPV-positive and HPV-negative cancers could lead to more personalized treatment approaches. Advances in immunotherapy, such as immune checkpoint inhibitors (e.g., PD-1/PD-L1 inhibitors), are showing promise in treating HPV-positive cancers due to the immune activation associated with HPV infection [27]. These therapies can enhance the ability of the immune system to recognize and attack cancer cells, leading to improved outcomes in HPV-positive cancers [28]. However, HPV-negative cancers, with their immune evasion mechanisms, may require alternative strategies, such as modulating the tumour microenvironment or targeting specific metabolic pathways [28]. The differential immune profiles between HPV-positive and HPV-negative cancers also open up opportunities for developing more effective biomarkers for early detection, prognosis, and treatment response [29].

The methodology of this study involved several key steps to analyze the relationship between HPV status and gene expression in CESC and HNSCC. Data was collected from The Cancer Genome Atlas (TCGA), including gene expression data and associated metadata such as patient demographics, clinical information, and HPV status. Pre-processing

the data involved cleaning to remove samples with missing or incomplete information, normalizing gene expression data using method log normalization, and filtering out low-expressing genes to focus on biologically significant ones. Differential gene expression analysis was performed to identify genes significantly upregulated or downregulated between HPV-positive and HPV-negative samples, with statistical significance determined by p-values < 0.05 and a log fold change cutoff of ± 2 , and false discovery rate correction was applied. An intersection of differentially expressed genes (DEGs) between both cancer types and HPV groups was analyzed to identify common genes with shared expression patterns. Pathway enrichment analysis was conducted using KEGG and Reactome to identify biological pathways associated with these DEGs, with significant pathways having p-values < 0.05. Metaflux analysis integrated gene expression data with metabolite flux data to explore metabolic variations between HPV-positive and HPV-negative samples. Deconvolution methods like CIBERSORT were used to estimate immune cell proportions from gene expression data to analyze the tumor microenvironment, comparing the immune cell composition between HPV-positive and HPV-negative samples. Finally, survival analysis was performed to assess the prognostic significance of specific genes using Kaplan–Meier survival curves and log-rank tests, identifying potential biomarkers associated with survival outcomes.

2 Methodology

2.1 Data collection

The data utilized in this study was collected from TCGA [30]. The data for CESC and HNSCC were retrieved. The dataset included both the primary gene expression data and the associated metadata, such as patient demographics, clinical information, HPV status (positive or negative), and other relevant biological annotations.

2.2 Data pre-processing

Several pre-processing steps were undertaken to ensure data quality and reliability for downstream analyses. Initially, data cleaning was performed to remove samples with incomplete metadata or missing critical information, ensuring consistency across datasets. To account for technical variability and enable cross-sample comparisons, gene expression data were normalized using Transcripts Per Million (TPM), a method chosen for its ability to correct for gene length differences while maintaining interpretability in transcript abundance. Low-expressing genes were filtered out to reduce noise and focus on biologically relevant signals. Batch effects, which can arise from differences in sequencing platforms or processing conditions, were assessed and corrected using ComBat, a robust empirical Bayes method, or linear regression models when appropriate. ComBat was selected for its effectiveness in adjusting for systematic technical variations without distorting biological differences, particularly in large-scale transcriptomic datasets. Quality control measures, including Principal Component Analysis (PCA), were applied to visualize data structure, confirm successful normalization, and identify potential outliers. These steps ensured that downstream analyses, such as differential expression and immune profiling, were performed on high-quality, standardized data.

2.3 Differential gene expression analysis

Following pre-processing, differential gene expression analysis was conducted to identify significantly upregulated or downregulated genes between HPV-positive and HPV-negative samples in both CESC and HNSCC. DESeq2 package was used to determine differential expression. Genes with a log fold change greater than 1 were considered upregulated, while those with a log fold change less than -1 were considered downregulated. A threshold of p-value < 0.05 was applied to identify statistically significant differentially expressed genes (DEGs) [31]. To control for type I errors, false discovery rate (FDR) correction was applied using the Benjamini–Hochberg method [32].

2.4 Identification of common genes

The intersection of differentially expressed genes was analyzed to elucidate shared molecular features. The lists of DEGs from HPV-positive and HPV-negative samples in both CESC and HNSCC were compared to identify common genes. These

genes were differentially expressed in both cancer types and shared between HPV-positive and HPV-negative groups and were considered for further analysis.

2.5 Pathway enrichment analysis

Pathway enrichment analysis was performed on differentially expressed genes (DEGs) to identify biological processes and signaling pathways significantly associated with HPV status. The analysis was conducted using Kyoto Encyclopedia of Genes and Genomes (KEGG) and Reactome databases. Gene set enrichment was evaluated using Gene Set Enrichment Analysis (GSEA) and over-representation analysis (ORA). GSEA was employed to determine whether predefined gene sets showed statistically significant differences between HPV-positive and HPV-negative samples, providing insights into coordinated gene expression changes. ORA was used to identify pathways disproportionately represented among DEGs. Enrichment scores were calculated using a weighted Kolmogorov–Smirnov statistic in GSEA, while statistical significance was assessed using permutation testing. Pathways with an adjusted p-value ($FDR < 0.05$) were considered significantly enriched. Functional pathway annotations were further validated using Enrichr to confirm key biological mechanisms underlying HPV-associated cancers [33].

2.6 Metaflux analysis

Metaflux analysis was conducted to examine metabolic pathway activity differences between HPV-positive and HPV-negative tumors. The analysis integrated transcriptomic data with metabolic flux modeling using the MetaFlux tool, which employs genome-scale metabolic models (GEMs) to infer pathway-level metabolic alterations. Flux balance analysis (FBA) was performed to estimate metabolic reaction rates under steady-state conditions. Differential metabolic pathway activities were identified using the Wilcoxon rank-sum test, and significantly altered pathways ($p < 0.05$) were mapped to KEGG metabolic networks. The analysis focused on key metabolic processes such as glycolysis, oxidative stress regulation, and amino acid metabolism, providing insights into metabolic adaptations driven by HPV status.

2.7 Tumor microenvironment analysis

To assess the tumor microenvironment composition, deconvolution analysis was performed using CIBERSORT, which estimates immune cell proportions based on gene expression profiles. The LM22 signature matrix, consisting of 22 immune cell types, was used to quantify immune cell infiltration in HPV-positive and HPV-negative tumors. The analysis employed a support vector regression algorithm to deconvolute bulk RNA-seq data into relative immune cell fractions. Comparisons between HPV status groups were performed using the Mann–Whitney U test, with p-values < 0.05 considered significant. Immune contexture differences, particularly the enrichment of CD8⁺ T cells in HPV-positive tumors and macrophage dominance in HPV-negative tumors were highlighted to provide mechanistic insights into HPV-associated immune responses [34].

2.8 Survival analysis

Survival analysis was conducted to evaluate the prognostic significance of specific genes involved in key pathways. Kaplan–Meier survival curves were generated to compare survival outcomes between high-expression and low-expression groups for selected genes. Statistical tests, such as log-rank tests, were applied to assess the significance of survival differences. This analysis identified potential prognostic markers and therapeutic targets linked to HPV-associated molecular mechanisms [35].

3 Results

3.1 Characteristics of data

The dataset analysed in this study comprised 166 samples for HNSCC and 62 samples for CESC and endocervical adenocarcinoma (CESC). For the HNSCC cohort, the dataset included 40 samples classified as normal, 84 samples from HPV-positive cases, and 42 samples from HPV-negative cases. Moreover, in the CESC cohort, the dataset contained 62 samples,

which included 3 samples classified as normal, 22 samples from HPV-positive cases, and 33 samples from HPV-negative cases.

3.2 Differentially expressed genes

To identify DEGs distinguishing tumor samples from normal samples in both HNSCC and CESC, we utilized the DESeq2 package in the R programming environment. This analysis applied stringent cutoffs based on p-values and Log2 (Fold Change). The Benjamini–Hochberg method was used to calculate adjusted p-values, with a significance threshold set at adjusted p-value < 0.05. For upregulated genes (activators), a Log2 (Fold Change) ≥ 1 was used as the cutoff, while for downregulated genes (suppressors), a Log2 (Fold Change) ≤ -1 was applied. In the HNSCC dataset, differential expression analysis identified 1958 genes associated with HPV-positive samples and 2151 genes associated with HPV-negative samples, showing significant differential expression compared to normal tissue. For the CESC dataset, 1117 DEGs were associated with HPV-positive samples, and 1034 DEGs were associated with HPV-negative samples (Fig. 1A, B). These genes were differentially expressed in both cancer types and shared between HPV-positive and HPV-negative groups and were considered for further analysis. Specifically, overlapping differentially expressed genes between HPV-positive and HPV-negative samples in HNSCC and CESC were retrieved. This analysis identified 149 overlapping genes associated with HPV-positive samples and 136 overlapping genes associated with HPV-negative samples (Fig. 2A, B). These shared genes represent critical molecular features that may be pivotal to understanding HPV-related oncogenesis.

3.3 Pathway analysis

The overlapping genes were subjected to KEGG pathway analysis to identify significant pathways. In HPV-negative samples, the major pathways included Viral protein interaction with cytokine and cytokine receptor, PPAR signaling pathway,

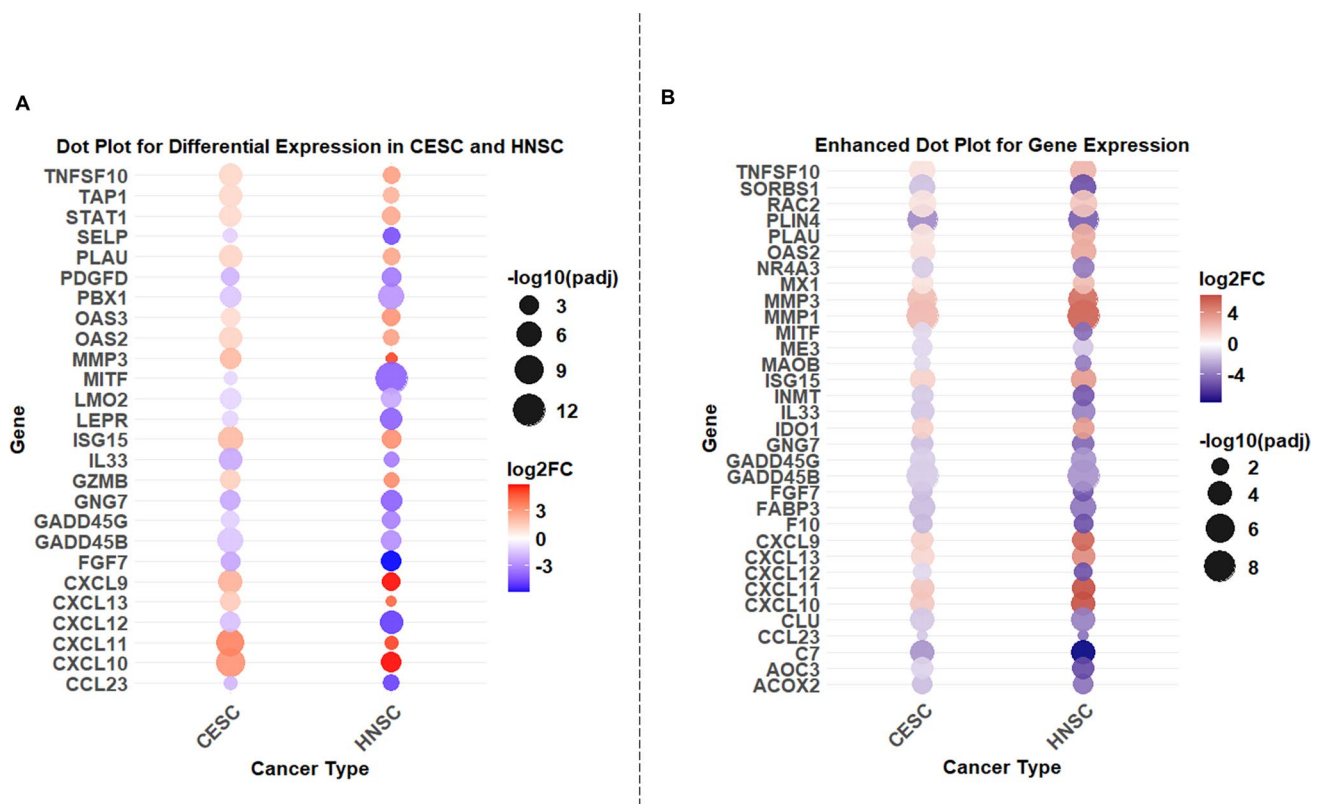


Fig. 1 **A** Scatter plot representing the common HPV-positive (HPV+) genes in cervical squamous cell carcinoma (CESC) and head and neck squamous cell carcinoma (HNSCC). The x-axis represents the log2 fold change ($\log_2\text{FC}$), while the y-axis represents the p-value. Blue dots correspond to genes from CESC, and red dots correspond to genes from HNSCC. **B** Scatter plot representing the common HPV-negative (HPV-) genes in CESC and HNSCC. The x-axis represents the log2 fold change ($\log_2\text{FC}$), while the y-axis represents the p-value. Blue dots correspond to genes from CESC, and red dots correspond to genes from HNSCC

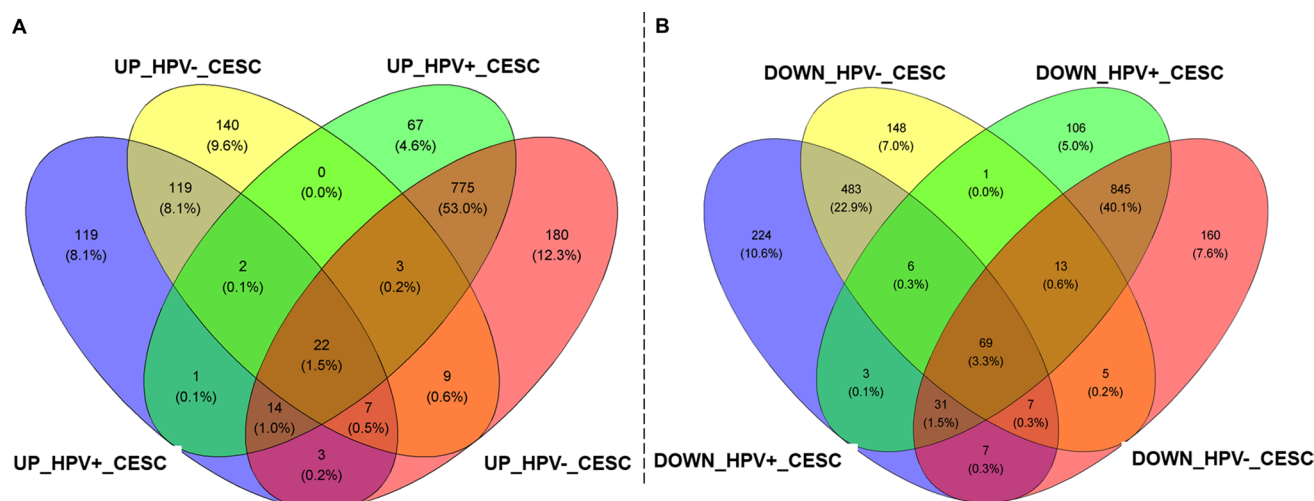


Fig. 2 **A** Venn diagram illustrating the common upregulated genes between HPV-positive (HPV+) and HPV-negative (HPV-) cases in head and neck squamous cell carcinoma (HNSCC) and cervical squamous cell carcinoma (CESC). The following colors depict the distribution of upregulated genes: yellow (upregulated genes unique to HPV- CESC), blue (upregulated genes unique to HPV+ CESC), green (upregulated genes unique to HPV+ HNSCC), and red (upregulated genes unique to HPV- HNSCC). Overlapping regions highlight the genes that are upregulated across multiple groups. **B** Venn diagram illustrating the common downregulated genes between HPV-positive (HPV+) and HPV-negative (HPV-) cases in HNSCC and CESC. The same color scheme represents the distribution of downregulated genes as Figure A: yellow (downregulated genes unique to HPV- CESC), blue (downregulated genes unique to HPV+ CESC), green (downregulated genes unique to HPV+ HNSCC), and red (downregulated genes unique to HPV- HNSCC). Overlapping regions indicate genes that are downregulated across multiple groups

Chemokine signaling pathway, Melanoma, Cytokine-cytokine receptor interaction, Coronavirus disease—COVID-19, Tryptophan metabolism, Transcriptional misregulation in cancer, Complement and coagulation cascades, and Phenylalanine metabolism. The major genes involved in major pathways include CXCL10, CXCL11, CXCL13, CXCL9, TNFSF10, CCL23, CXCL12, MMP1, ACOX2, FABP3, ME3, PLIN4, SORBS1, RAC2, GNG7, FGF7, GADD45B, GADD45G, MITF, IL33, ISG15, MMP3, MX1, OAS2, C7, IDO1, INMT, MAOB, PLAUI, NR4A3, CLU, F10, AOC3 (Fig. 3A). For HPV-positive samples, the major enriched pathways included Viral protein interaction with cytokine and cytokine receptor, Chemokine signaling pathway, Transcriptional misregulation in cancer, Epstein-Barr virus infection, Melanoma, Cytokine-cytokine receptor interaction, Influenza A, Coronavirus disease—COVID-19, NF-kappa B signaling pathway, and Toll-like receptor signaling pathway. The

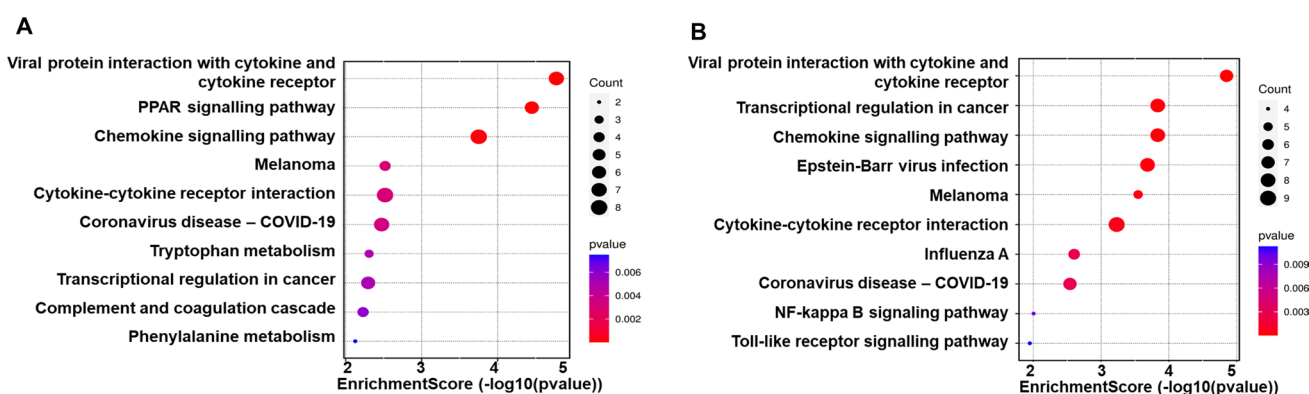


Fig. 3 **A** Presents a dot plot of pathways associated with HPV-negative (HPV-) genes in CESC and HNSCC, following the same conventions for p-value and gene count representation. These plots highlight the differential enrichment of pathways in HPV+ and HPV- gene sets, providing a comprehensive overview of the functional roles and biological processes significantly influenced by HPV status across these cancer types. **B** Depicts a dot plot illustrating the major pathways associated with HPV-positive (HPV+) genes in cervical squamous cell carcinoma (CESC) and head and neck squamous cell carcinoma (HNSCC). The pathways are visualized with a dual representation of significance and gene count. The color gradient of the dots indicates the p-value, with blue representing the most significant pathways and transitioning to red for less significant pathways. The size of the dots reflects the number of genes associated with each pathway, where larger dots correspond to a higher number of genes smaller dots indicate fewer genes

major genes involved in major pathways include CXCL10, STAT1, CXCL11, CXCL9, CXCL12, GADD45B, GADD45G, CXCL13, TNFSF10, CCL23, OAS2, OAS3, MMP3, PLA2, MITF, ISG15, IL33, GNG7, GZMB, LMO2, PBX1, TAP1, FGF7, PDGFD, LEPR, SELP (Fig. 3B). All pathways were statistically significant, with p-values less than 0.005, showing their potential involvement in HPV-associated molecular mechanisms.

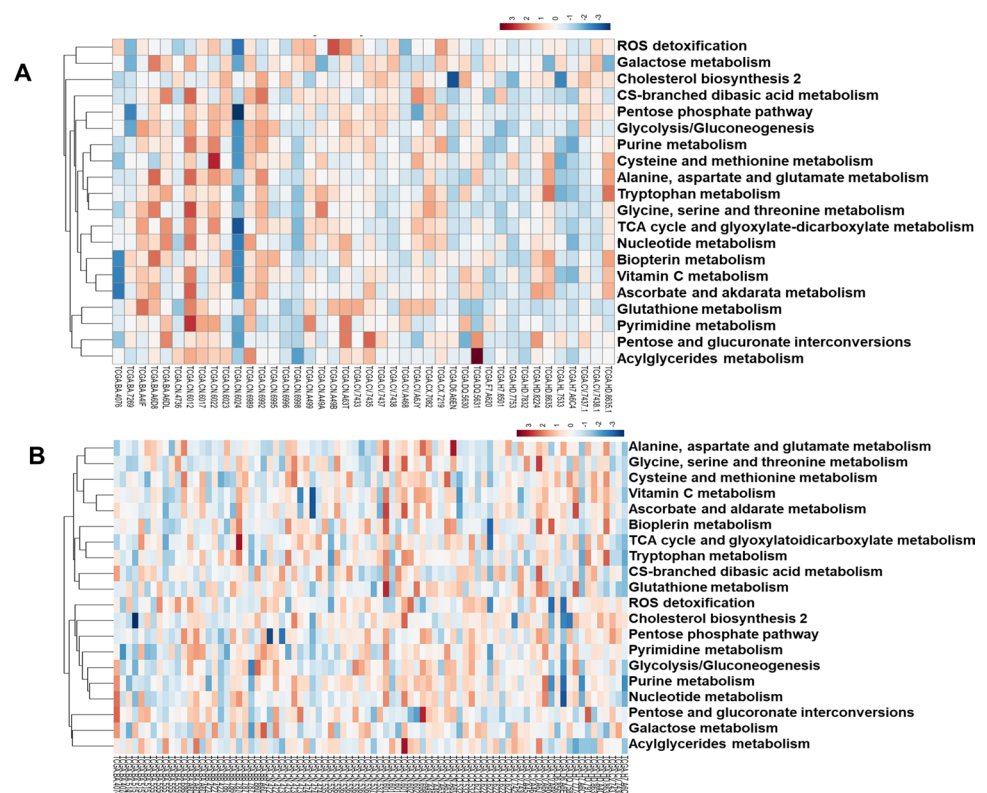
3.4 Metaflux analysis

The metaflux analysis revealed complex relationships between metabolites and the samples, highlighting significant variations in associated pathways despite no major differences between positive and negative fluxes. In HPV-positive samples, pathways such as ROS detoxification, galactose metabolism, cholesterol biosynthesis 2, CS-branched dibasic acid metabolism, the pentose phosphate pathway, glycolysis/gluconeogenesis, purine metabolism, cysteine and methionine metabolism, alanine, aspartate, and glutamate metabolism, tryptophan metabolism, glycine, serine, and threonine metabolism, the tricarboxylic acid cycle and glyoxylate-dicarboxylate metabolism, nucleotide metabolism, bipterin metabolism, vitamin C metabolism, ascorbate and aldarate metabolism, glutathione metabolism, pyrimidine metabolism, and pentose and glucuronate interconversions were predominant. Conversely, HPV-negative samples were enriched in overlapping pathways, including alanine, aspartate, and glutamate metabolism; glycine, serine, and threonine metabolism; cysteine and methionine metabolism; vitamin C metabolism; ascorbate and aldarate metabolism; bipterin metabolism; the tricarboxylic acid cycle and glyoxylate-dicarboxylate metabolism; tryptophan metabolism; CS-branched dibasic acid metabolism; glutathione metabolism; ROS detoxification; cholesterol biosynthesis 2; the pentose phosphate pathway; pyrimidine metabolism; glycolysis/gluconeogenesis; purine metabolism; nucleotide metabolism; pentose and glucuronate interconversions; galactose metabolism; and acylglycerides metabolism, reflecting nuanced metabolic distinctions between the two sample types (Fig. 4A, B).

3.5 Survival analysis

Survival analysis was conducted to evaluate the prognostic significance of genes involved in major pathways. Specifically, CXCL11 was analyzed in HPV-positive HNSCC and CESC, and STAT1 was analyzed in HPV-negative HNSCC and CESC. Kaplan–Meier survival curves were generated, and survival differences were assessed between high-expression and

Fig. 4 **A** presents a heatmap illustrating the relationship between HPV-positive (HPV+) samples and their associated metabolic pathways. The y-axis displays the metabolic pathways, while the x-axis represents the HPV+ samples. The heatmap employs a color-coded scheme where red shades indicate positive correlations between samples and pathways, and blue shades represent negative correlations. This visualization captures the distinct metabolic profiles linked to HPV+ status. **B** similarly showcases a heatmap for HPV-negative (HPV-) samples, adhering to the same format and color-coding scheme



low-expression groups for these genes. The results indicated that low expression of both CXCL11 and STAT1 was associated with reduced survival compared to their high-expression counterparts in both cancers. Notably, CXCL11, which is predominantly associated with HPV-positive cases, showed a greater reduction in survival compared to STAT1 across both cancer types. This suggests a distinct impact of these genes on patient outcomes, emphasizing their potential as therapeutic targets or prognostic markers (Fig. 5A, D).

3.6 Deconvolution analysis

Deconvolution analysis of the tumor microenvironment was conducted using CIBERSORT to estimate immune cell proportions in HPV-positive and HPV-negative tumors. In the HNSCC cohort, HPV-positive samples displayed a dual association: they correlated with both macrophages and CD8+ memory cells, each with correlation coefficients greater than 0.3, indicating a complex immune landscape. HPV-negative HNSCC tumors were primarily macrophage-enriched, with correlation coefficients exceeding 0.3, reinforcing their immune-suppressive nature (Fig. 6A, C). In the CESC cohort, HPV-positive samples exhibited higher CD8+T cell infiltration, with correlation coefficients around 0.4–0.5, suggesting

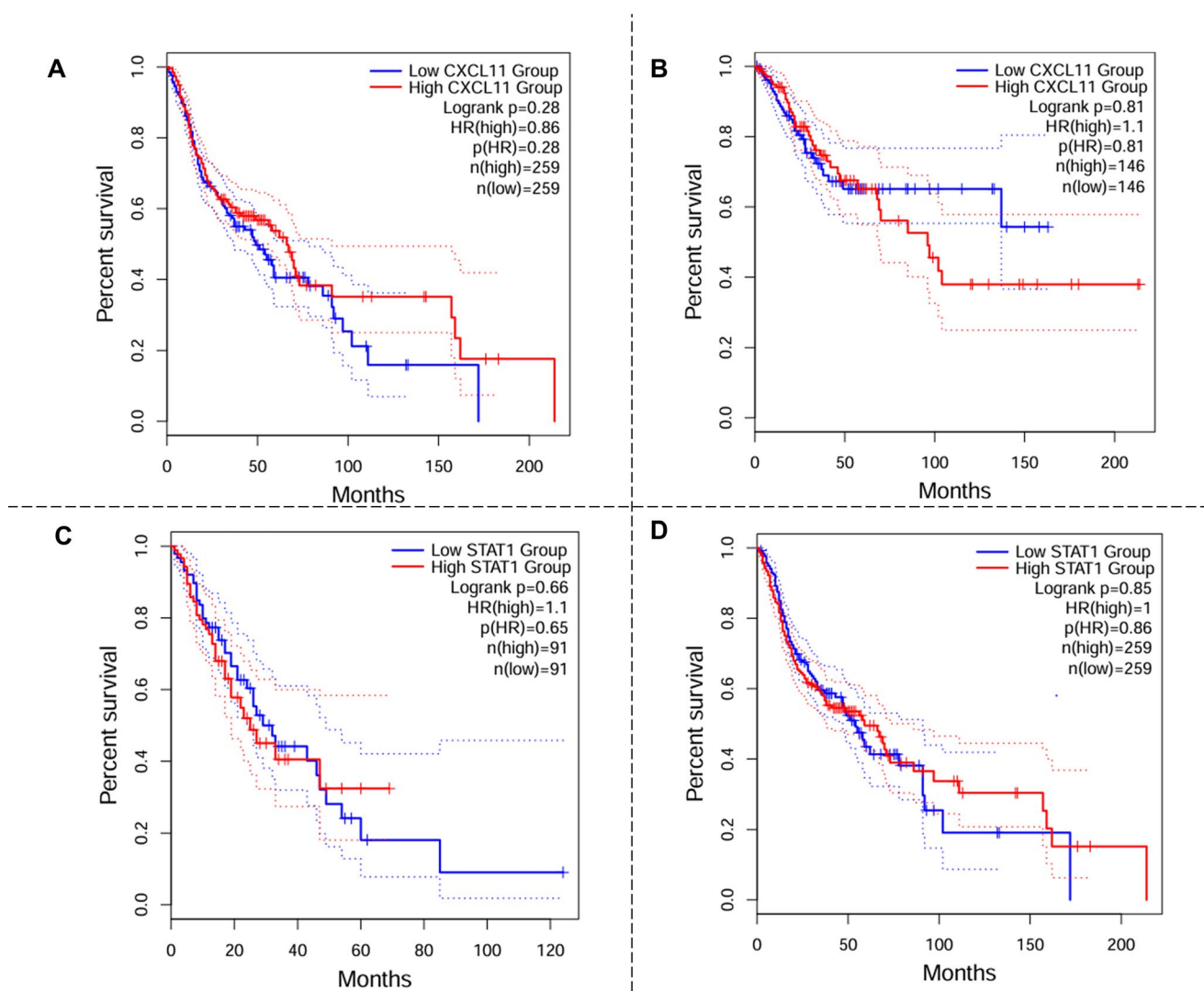


Fig. 5 A–D The panels illustrate the overall survival rates associated with CXCL11 and STAT1 expression in HNSCC and CESC, respectively. In each survival plot, the y-axis represents the survival percentage, while the x-axis denotes the months of survival. The red curve corresponds to the high-expression group, indicating patients with elevated levels of the respective gene, whereas the blue curve represents the low-expression group. These survival analyses highlight the potential prognostic significance of CXCL11 and STAT1 expression in HPV-related cancers

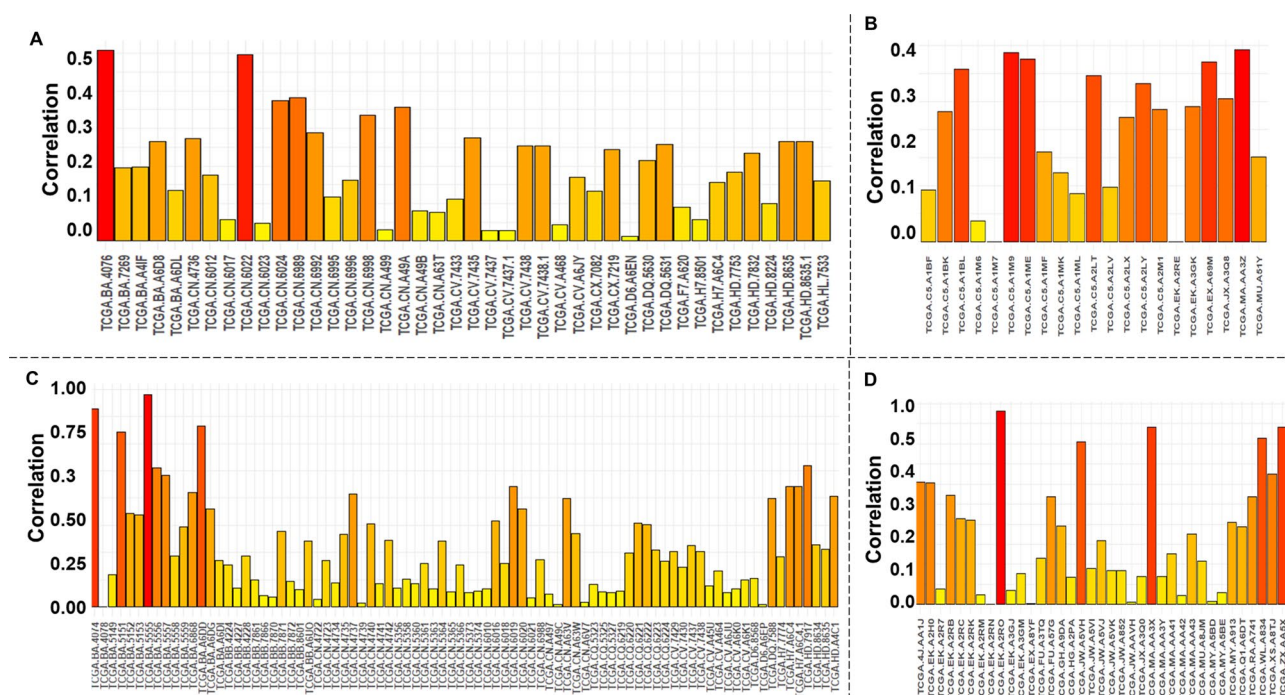


Fig. 6 A–D Correlation between immune cell infiltration levels and HPV status in HNSCC and CESC. **A,B** CD8+T cell infiltration correlation: HPV-positive HNSC and CESC tumors exhibit significantly higher CD8+T cell infiltration, with correlation coefficients ranging between 0.4 and 0.5, highlighting a more immune-active tumor microenvironment. HPV-negative samples show lower CD8+T cell correlation, indicating reduced cytotoxic immune response. **C,D** Macrophage infiltration correlation: HPV-negative HNSC and CESC tumors are predominantly macrophage-enriched, with correlation coefficients exceeding 0.3–0.4, suggesting a more immunosuppressive phenotype. HPV-positive tumors show moderate to weak correlation with macrophages, indicating distinct immune microenvironmental differences driven by HPV status. The y-axis represents the Spearman correlation coefficient, suggesting the association strength between immune cell infiltration and tumor status. The x-axis represents the number of analyzed samples. Red dots depict individual sample correlation values. These findings emphasize the distinct immune landscapes of HPV-driven versus non-HPV-driven cancers and suggest potential immune-targeted therapeutic implications

a more immunologically active profile. In contrast, HPV-negative CESC tumors were predominantly associated with macrophages, with a correlation coefficient exceeding 0.4, indicating an immunosuppressive microenvironment (Fig. 6B, D). These immune infiltration patterns highlight the therapeutic implications of HPV-driven versus non-HPV-driven tumors, suggesting immune checkpoint inhibitors as a potential strategy for HPV-positive cases while macrophage-targeting therapies might be more effective in HPV-negative tumors.

4 Discussion

HPV plays a significant role in the pathogenesis of both HNSCC and CESC, though its impact and the mechanisms it influences these cancers differ [36]. In HNSCC, particularly oropharyngeal cancers, HPV infection is a major etiological factor, with HPV-positive tumors exhibiting distinct clinical and molecular characteristics, such as improved prognosis compared to HPV-negative cases [37]. In contrast, HPV is well-established as a primary cause of CESC, where persistent infection can lead to the development of pre-cancerous lesions and invasive carcinoma [18]. Understanding the differences between HPV-associated HNSCC and CESC is crucial, as these cancers arise in different anatomical sites, have varying responses to treatment, and exhibit different immune microenvironments. The distinction is important for tailoring therapeutic strategies, improving early detection methods, and developing HPV-targeted interventions. As the molecular profiles and clinical outcomes of these cancers vary significantly based on HPV status, research into the specific role of HPV in each cancer type can contribute to better prognostic tools and personalized treatment approaches.

This study employed advanced bioinformatics tools to explore the dynamics of HPV-positive and HPV-negative CESC and HNSCC, revealing differences in gene expression, metabolic activity, immune composition, and survival outcomes between these two groups. Pathway enrichment analysis revealed distinct biological pathways enriched in HPV-positive

and HPV-negative samples in both CESC and HNSCC. Several pathways related to immune modulation and viral interactions in HPV-positive samples, such as the cytokine-cytokine receptor interaction and viral protein interaction with cytokines, were significantly enriched. These findings align with previous studies suggesting that HPV infection triggers an inflammatory immune response and activates host pathways that may contribute to cancer development [38]. The significant enrichment in these pathways suggests that HPV-associated cancers might utilize immune signaling pathways to promote tumorigenesis. On the other hand, HPV-negative samples exhibited enrichment in pathways related to cell metabolism, such as the PPAR signaling pathway, melanoma, and transcriptional misregulation in cancer. These pathways are important to developing cancers that may not rely on viral interactions but exploit dysregulated cellular processes such as metabolism and immune escape. The findings from this analysis show the fundamental differences at the molecular level between HPV-positive and HPV-negative cancers, suggesting that HPV status could inform the design of targeted therapies and biomarkers. The pathway analysis also revealed the overlap in some pathways between both HPV-positive and HPV-negative cancers, particularly those involved in immune response regulation. These shared pathways might indicate a common immune-related mechanism in both types of cancers, suggesting that immune modulation could be a promising therapeutic approach in HPV-positive and HPV-negative cancers alike. Metaflux analysis provided crucial information into the metabolic variations associated with HPV status. HPV-positive samples were found to be enriched in pathways related to ROS (Reactive Oxygen Species) detoxification, galactose metabolism, cholesterol biosynthesis, and glycolysis. These pathways indicate metabolic adaptations required to support the high proliferative rates characteristic of cancer cells. ROS detoxification, for instance, suggests that HPV-positive tumors may exhibit a higher level of oxidative stress, necessitating mechanisms to mitigate cellular damage. The increased reliance on glycolysis (Warburg effect) is a well-known characteristic of cancer cells, and its presence in HPV-positive samples suggests that HPV infection could promote the metabolic shift towards glycolysis to sustain energy production in rapidly dividing cells. In contrast, HPV-negative samples showed significant enrichment in metabolic pathways such as glycine, serine, threonine, vitamin C, and pyrimidine metabolism. These findings point to a different set of metabolic demands in HPV-negative cancers. The involvement of these pathways may reflect a reliance on amino acid and nucleotide metabolism, which are crucial for supporting cellular growth and DNA synthesis in the absence of viral oncogenesis. Understanding these distinct metabolic pathways offers potential therapeutic opportunities, such as targeting specific metabolic vulnerabilities in HPV-positive or HPV-negative tumours. Moreover, the differences in metabolic flux between HPV-positive and HPV-negative cancers highlight the potential of using metabolic inhibitors or therapies targeting specific metabolic enzymes as a personalized treatment strategy based on HPV status. This finding aligns with the growing literature exploring metabolic reprogramming in cancer as a promising therapeutic approach. The tumor microenvironment (TME) plays a crucial role in cancer progression, influencing both tumor cell behavior and the immune response. In this study, deconvolution analysis was performed to assess the immune cell composition within the TME. HPV-positive samples in the CESC cohort showed a significant correlation with CD8 + naïve T cells, suggesting a robust adaptive immune response. In contrast, HPV-negative CESC samples were predominantly associated with macrophages, indicating a shift toward immune suppression or evasion. This discrepancy in immune cell composition suggests that HPV-positive cancers may have a more active immune response, which could potentially be leveraged for immunotherapy. Similarly, in the HNSCC cohort, HPV-positive samples demonstrated correlations with both macrophages and CD8 + memory T cells, whereas HPV-negative samples were predominantly associated with macrophages. This finding suggests that HPV-negative HNSCC might exploit macrophage-mediated immune suppression, further complicating the immune landscape in these cancers. These results align with the growing recognition that macrophages, particularly tumor-associated macrophages (TAMs), play a critical role in the immunosuppressive microenvironment of many cancers. The immune composition differences in the TME between HPV-positive and HPV-negative cancers have important implications for immunotherapy [39]. HPV-positive cancers, with their association with active CD8 + T cell responses, may benefit more from immune checkpoint inhibitors targeting PD-1/PD-L1 or CTLA-4. In contrast, HPV-negative cancers, with a stronger macrophage presence, may require therapies aimed at modulating the TME to enhance anti-tumor immunity. Survival analysis provided a deeper understanding of the prognostic value of key genes involved in HPV-related cancers. Specifically, the analysis focused on genes like CXCL11 in HPV-positive cases and STAT1 in HPV-negative cancers. Kaplan–Meier survival curves demonstrated that low expression of CXCL11 and STAT1 was associated with poor survival outcomes in both cancer types, with CXCL11 showing a more pronounced effect in HPV-positive cases. CXCL11, a chemokine associated with immune cell recruitment, is known to regulate T-cell responses [40]. Its low expression in HPV-positive cancers may indicate an impaired immune response, leading to a worse prognosis. Similarly, STAT1, a critical transcription factor in immune response regulation, showed reduced survival in low-expression HPV-negative cancers. This suggests that impaired immune signaling through pathways regulated by STAT1 could contribute to poor prognosis in these cancers. These

findings underscore the potential of CXCL11 and STAT1 as biomarkers for prognosis in HPV-associated cancers. Their role in regulating immune responses makes them promising candidates for targeted therapies to modulate the immune microenvironment in both HPV-positive and HPV-negative cancers. Integrating pathway enrichment analysis, metaflux analysis, TME composition, and survival analysis offers a comprehensive view of the molecular landscape of HPV-positive and HPV-negative cancers. The pathway and metabolic differences between HPV-positive and HPV-negative cancers reveal that these two subtypes rely on distinct molecular and metabolic pathways for survival and proliferation. HPV-positive cancers exhibit a greater dependence on immune signaling and oxidative stress regulation, while HPV-negative cancers rely more on metabolic processes like amino acid metabolism and nucleotide synthesis. Furthermore, the immune composition differences between HPV-positive and HPV-negative samples suggest that HPV status significantly influences the tumor microenvironment. HPV-positive cancers are associated with a more robust immune response, potentially offering therapeutic opportunities with immune checkpoint inhibitors. In contrast, HPV-negative cancers may require strategies to alter the immune landscape and reduce macrophage-mediated suppression. The survival analysis results further reinforce the importance of immune-related genes like CXCL11 and STAT1 in cancer prognosis. These genes may serve as valuable prognostic markers and therapeutic targets, highlighting the need for personalized treatment approaches based on HPV status and immune microenvironment characteristics. This study comprehensively analyses the molecular and immune landscape of HPV-positive and HPV-negative cancers, identifying key pathways, metabolic alterations, and immune cell compositions that differentiate these subtypes. The results emphasize the importance of HPV status in determining the biological characteristics and therapeutic strategies for cervical and head and neck cancers. The findings suggest that HPV-positive cancers may benefit from immune-based therapies targeting immune signaling pathways. In contrast, HPV-negative cancers may require approaches focused on metabolic reprogramming and modulation of the tumor microenvironment. Integrating these molecular and immune insights could pave the way for more effective, personalized treatment strategies for HPV-associated cancers.

Our findings highlight distinct molecular and immunological differences between HPV-positive and HPV-negative cancers, emphasizing the divergent tumor microenvironments and their therapeutic implications. Specifically, our data suggest that HPV-positive tumors exhibit a greater dependence on immune signaling pathways, as evidenced by increased immune cell infiltration and upregulation of immune-related genes. This observation aligns with previous studies indicating that the presence of viral antigens in HPV-positive tumors elicits a stronger immune response, potentially explaining the observed clinical benefit of immune checkpoint inhibitors (ICIs) in these patients. For instance, Cha et al. (2024) demonstrated that HPV-positive head and neck squamous cell carcinomas (HNSCCs) are characterized by an inflamed immune microenvironment with elevated levels of CD8⁺ T cells and PD-1 expression, correlating with improved responses to ICIs [41]. However, the extent of immune activation in HPV-positive cancers may vary across cohorts due to multiple factors, including tumor heterogeneity, host immune status, differences in HPV subtype prevalence, and variations in sequencing methodologies. For example, while some studies have reported robust immune infiltration in HPV-positive tumors, others suggest that immune evasion mechanisms, such as PD-L1 overexpression, T-cell exhaustion, and regulatory T-cell enrichment, may modulate the overall response to immunotherapy. Additionally, the interplay between the immune microenvironment and co-occurring genetic alterations, such as TP53 and PIK3CA mutations, may further influence immune activation and therapeutic outcomes [42].

Conversely, our analysis indicates that HPV-negative cancers exhibit a relative lack of immune infiltration and appear to rely more on metabolic and proliferative pathways. This suggests that while immune-based therapies may have limited efficacy in HPV-negative tumors, targeting metabolic vulnerabilities could offer alternative therapeutic strategies. Dysregulated metabolic pathways, such as enhanced glycolysis, lipid metabolism, and oxidative phosphorylation, have been implicated in immune evasion and tumor progression in HPV-negative cancers. For instance, a study by Li et al. (2024) found that HPV-negative HNSCCs display increased glycolytic activity and reduced mitochondrial respiration compared to HPV-positive counterparts, highlighting potential metabolic targets for therapy [43].

A key challenge in translating our findings into clinical applications is identifying practical implementation strategies for targeted therapeutic approaches based on HPV status. While our study highlights distinct immune and metabolic dependencies in HPV-positive and HPV-negative cancers, further validation in preclinical models and clinical trials is necessary to refine these insights into actionable interventions. Several ongoing clinical trials are already evaluating immune checkpoint inhibitors (ICIs) in HPV-positive cancers [44, 45], highlighting the potential for stratified immunotherapy approaches. However, variability in response rates suggests that additional biomarker-driven patient selection strategies, such as T-cell infiltration profiles and PD-L1 expression levels, must be incorporated into future trial designs.

For HPV-negative tumors, the identification of metabolic vulnerabilities suggests an opportunity for therapeutic intervention through metabolic inhibitors. Preclinical studies have demonstrated that targeting glycolysis (e.g., using

2-deoxy-D-glucose) or mitochondrial oxidative phosphorylation (OXPHOS) inhibitors can sensitize HPV-negative tumors to existing therapies. However, these findings require further clinical validation through trials integrating metabolic targeting strategies with standard-of-care treatments such as radiation or chemotherapy [46, 43]. Future research should focus on conducting patient-derived xenograft (PDX) and organoid models to validate these metabolic dependencies and assess their therapeutic relevance in clinically diverse cohorts.

Future studies integrating metabolomics, functional assays, and spatial transcriptomics will also be crucial in identifying specific metabolic dependencies that can be exploited for targeted therapy.

5 Limitations of the study

While this study provides valuable insights into the distinct metabolic and immune profiles of HPV-positive and HPV-negative head and neck squamous cell carcinoma (HNSCC) and cervical squamous cell carcinoma (CESC), a few limitations exist. Future studies incorporating experimental validation, longitudinal data, and patient-derived samples will be critical for refining these findings and advancing their translational potential.

1. The study heavily relies on publicly available datasets, primarily from TCGA, which may introduce biases related to sample collection, processing, and demographic representation.
2. Although bioinformatics tools were extensively utilized to analyze gene expression, metabolic pathways, and immune profiles, experimental validation of key findings, such as the role of specific metabolic pathways or immune markers in HPV-related cancers, was not performed.
3. The immune cell composition was inferred using computational algorithms, which provide a static snapshot of immune infiltration. Dynamic interactions within the tumor microenvironment, such as temporal changes in immune cell recruitment or activation, were not explored.
4. While CXCL11 and STAT1 were identified as potential prognostic markers, the study does not comprehensively assess other biomarkers that could influence prognosis or therapeutic response in HPV-related cancers.
5. Metaflux analysis provides predictions of metabolic activity based on gene expression data, but these predictions do not directly measure metabolic fluxes or account for post-transcriptional and post-translational regulation.

6 Conclusion

In conclusion, this study emphasizes the significant role of HPV in shaping the molecular characteristics and clinical outcomes of both HNSCC and CESC. By identifying differentially expressed genes, biological pathways, and immune microenvironment variations associated with HPV status, the research provides essential insights into the unique mechanisms underlying these cancers. Additionally, the exploration of metabolic pathways revealed distinct metabolic shifts linked to HPV status, highlighting how HPV influences cancer metabolism and potentially offers new therapeutic targets. Understanding these metabolic alterations is crucial for developing novel treatment strategies that could disrupt cancer cell metabolism. Future research should focus on expanding these findings by further investigating the functional roles of identified biomarkers, the impact of metabolic reprogramming, and the potential of HPV-targeted therapies. Integrating multi-omics data, including metabolomics, could provide a deeper understanding of tumor biology and metabolic adaptations in both HNSCC and CESC. Prospective clinical trials focusing on HPV-specific therapies and metabolic interventions could significantly improve the management of these cancers, offering innovative therapeutic options and enhanced patient outcomes.

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Data availability Data is provided within the manuscript.

Declarations

Ethics approval and consent to participate Not applicable.

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